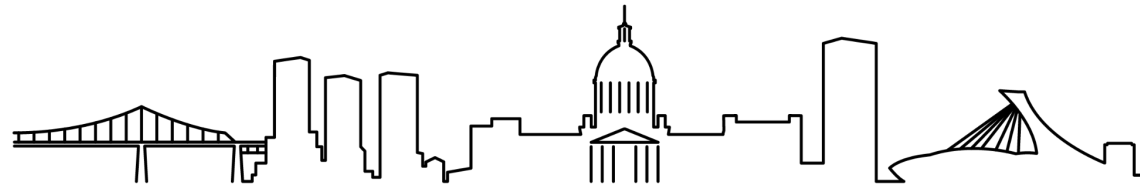


Hands-on 3 Segmentation

(Keras3/Tensorflow environnement)



— Montreal, June 15–19 —

DLM I 2026

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Preliminary remarks

- All four hands-on can be done on a computing cloud such as that of *CalculQuebec*, *Google Colab*, ... or on your own computer
- **For beginners** : we recommend you to start with *CalculQuebec*
- For advanced users with CPU/GPU access
 - You can run on your specific hardware
 - Tips for installing the required environnement are [here](#).

First, setup your *CalculQuebec* Account

(cf. DLMI 2026 hands on setup instructions)

1. *Create an account on CalculQuebec Cloud and log-in*
<https://jupyter.dmlri.calculquebec.cloud/hub/login>
2. Create a compute node with
 1. Memory at 4096MB
 2. GPU 1x1G.5GB
3. Start the jupyter-lab server and wait for it **Steps covered here**
4. Download the hands-on material into your jupyter-lab
5. Open a notebook
6. Cells philosophy and execution

Jupyter Lab interface (can change)

The screenshot shows the Jupyter Lab web interface. On the left is a file browser sidebar with a table of files and folders. Two orange arrows point to the 'upload' button (a square with an upward arrow) and the file list area. A red text box at the bottom left says "Drag and drop files here (or via the upload button)". The main area is titled "Launcher" and contains several tabs: "Notebook", "Console", "Other", and "Documentation". A green arrow points to the "Launcher tab" label. Another green arrow points to the "Terminal" icon in the "Other" section. The "Notebook" section shows icons for Python 3.11, LibreQDA, Mate Desktop (VNC), MLflow, OpenRefine, Python bundle, RStudio, Tensorboard, and VS Code. The "Console" section shows icons for Python 3.11 and Python bundle. The "Other" section shows icons for Terminal, Job statistics, Text File, Markdown File, Python File, and Show Contextual Help. The "Documentation" section shows icons for General and Jupyter. A small notification box in the bottom right corner says "Would you like Jupyter news".

Drag and drop files here
(or via the upload button)

Launcher tab

Terminal

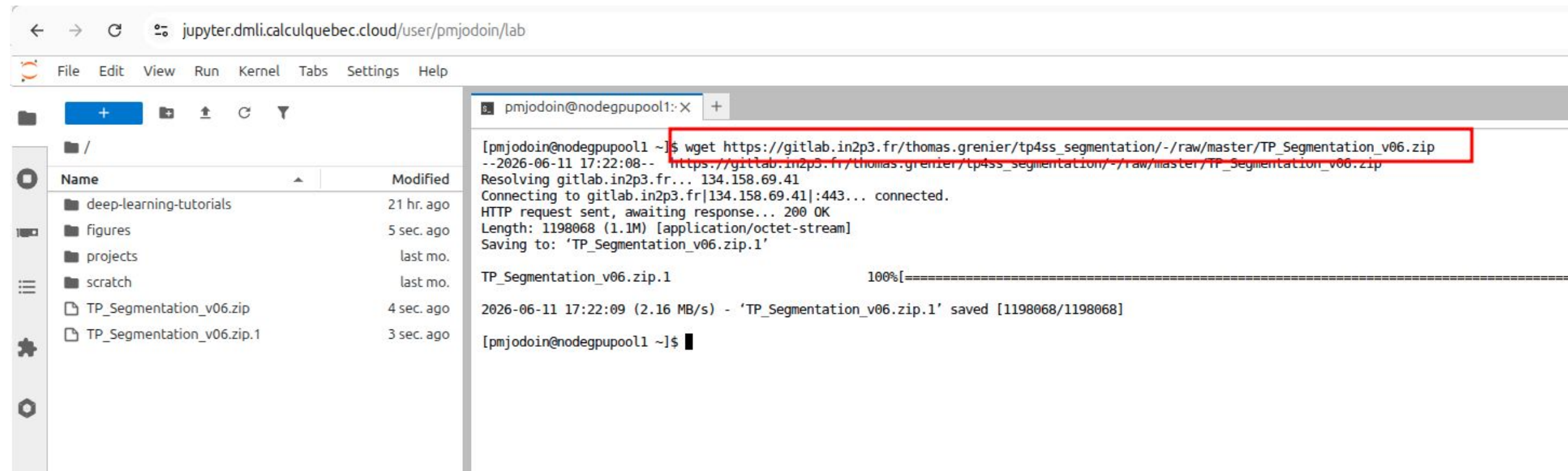
Download Hands-on materials

(same as for hands-on 1 and 2)

a- Open a **Terminal** (c.f. previous page)

b- download from the terminal with the following command :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp4ss_segmentation/-/raw/master/TP_Segmentation_v06.zip
```



The screenshot displays the JupyterLab web interface. On the left, a file browser shows a directory structure with folders like 'deep-learning-tutorials', 'figures', 'projects', and 'scratch', and files 'TP_Segmentation_v06.zip' and 'TP_Segmentation_v06.zip.1'. The main area on the right contains a terminal window titled 'pmjodoin@nodegpupool1: X'. The terminal shows the execution of the command `wget https://gitlab.in2p3.fr/thomas.grenier/tp4ss_segmentation/-/raw/master/TP_Segmentation_v06.zip`, which is highlighted with a red box. The output of the command is as follows:

```
[pmjodoin@nodegpupool1 ~]$ wget https://gitlab.in2p3.fr/thomas.grenier/tp4ss_segmentation/-/raw/master/TP_Segmentation_v06.zip
--2026-06-11 17:22:08-- https://gitlab.in2p3.fr/thomas.grenier/tp4ss_segmentation/-/raw/master/TP_Segmentation_v06.zip
Resolving gitlab.in2p3.fr... 134.158.69.41
Connecting to gitlab.in2p3.fr|134.158.69.41|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1198068 (1.1M) [application/octet-stream]
Saving to: 'TP_Segmentation_v06.zip.1'

TP_Segmentation_v06.zip.1          100%[=====]
2026-06-11 17:22:09 (2.16 MB/s) - 'TP_Segmentation_v06.zip.1' saved [1198068/1198068]

[pmjodoin@nodegpupool1 ~]$
```

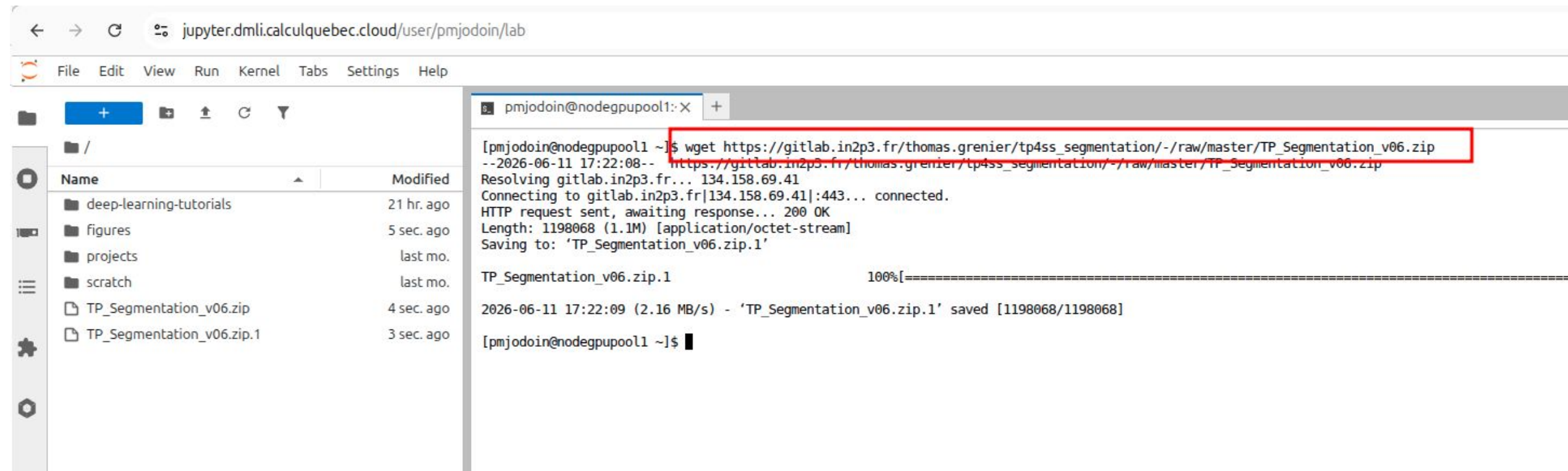
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```
[pmjodoin@nodegpupool1 ~]$ wget https://gitlab.in2p3.fr/thomas.grenier/tp4ss_segmentation/-/raw/master/TP_Segmentation_v06.zip
--2026-06-11 17:22:08-- https://gitlab.in2p3.fr/thomas.grenier/tp4ss_segmentation/-/raw/master/TP_Segmentation_v06.zip
Resolving gitlab.in2p3.fr... 134.158.69.41
Connecting to gitlab.in2p3.fr|134.158.69.41|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1198068 (1.1M) [application/octet-stream]
Saving to: 'TP_Segmentation_v06.zip.1'

TP_Segmentation_v06.zip.1      100%[=====]
2026-06-11 17:22:09 (2.16 MB/s) - 'TP_Segmentation_v06.zip.1' saved [1198068/1198068]

[pmjodoin@nodegpupool1 ~]$
```

Download Hands-on materials

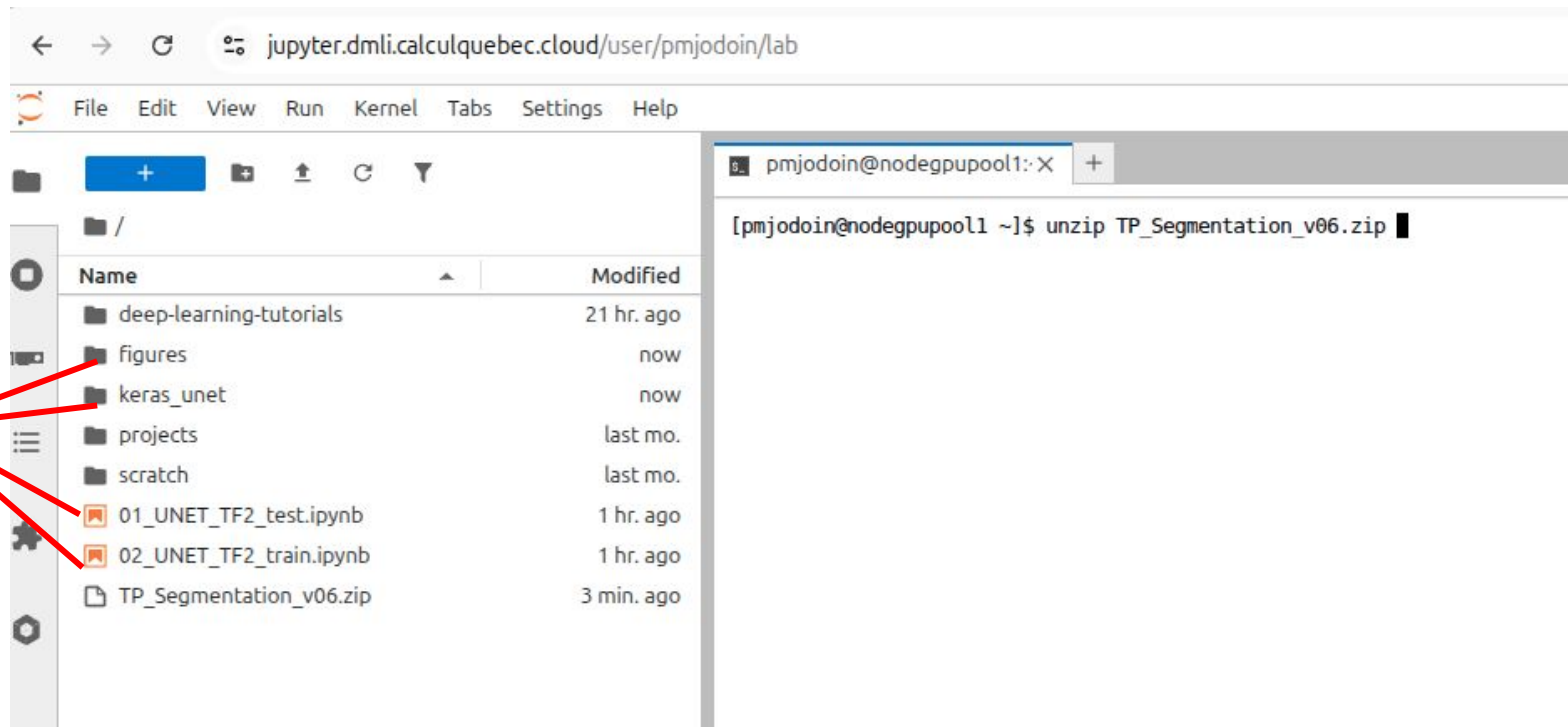
(same as for hands-on 1 and 2)

a- Open a **Terminal** (c.f. previous page)

b- download from the terminal with the following command :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp4ss_segmentation/-/raw/master/TP_Segmentation_v06.zip
```

c- unzip from jupyter lab (in a Terminal)



unzipped files

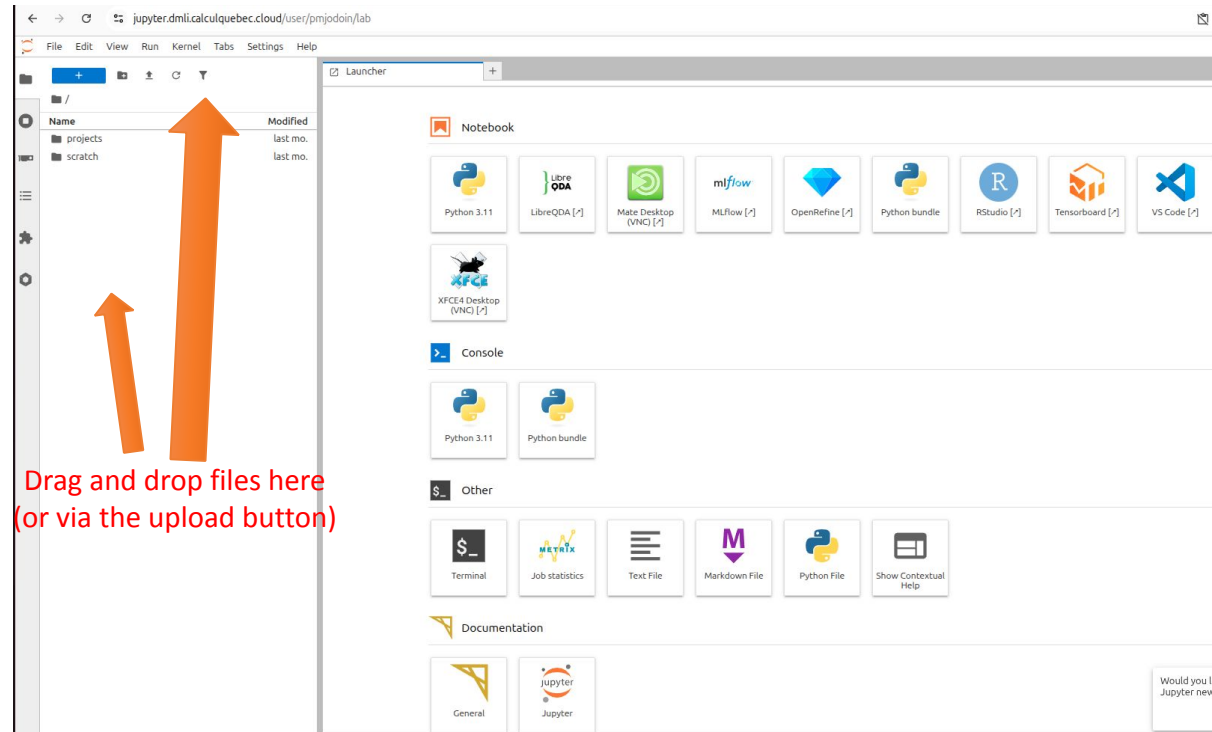
Download Hands-on materials

If the previous step did not work, you can

a) download the zip file on your computer

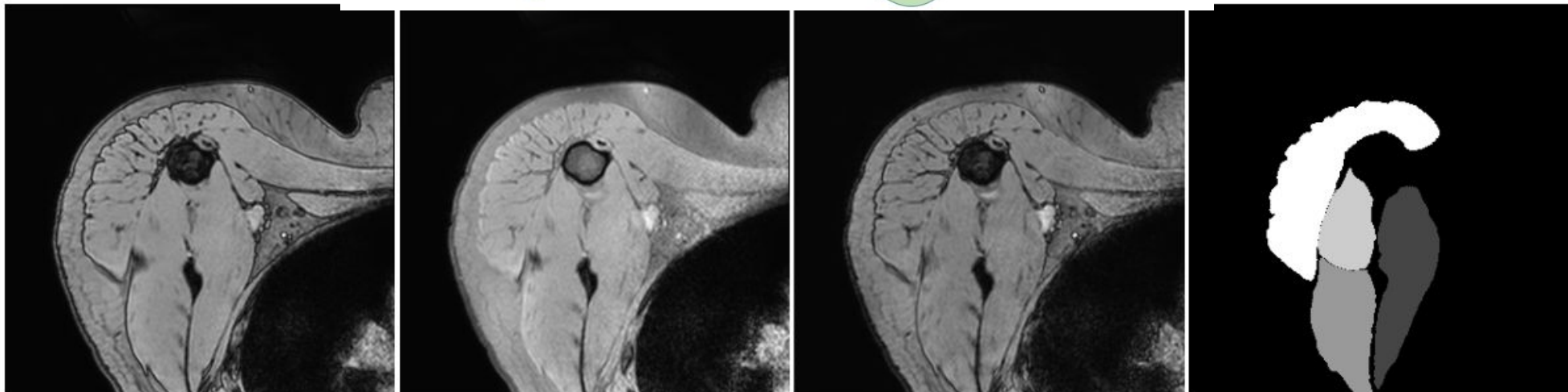
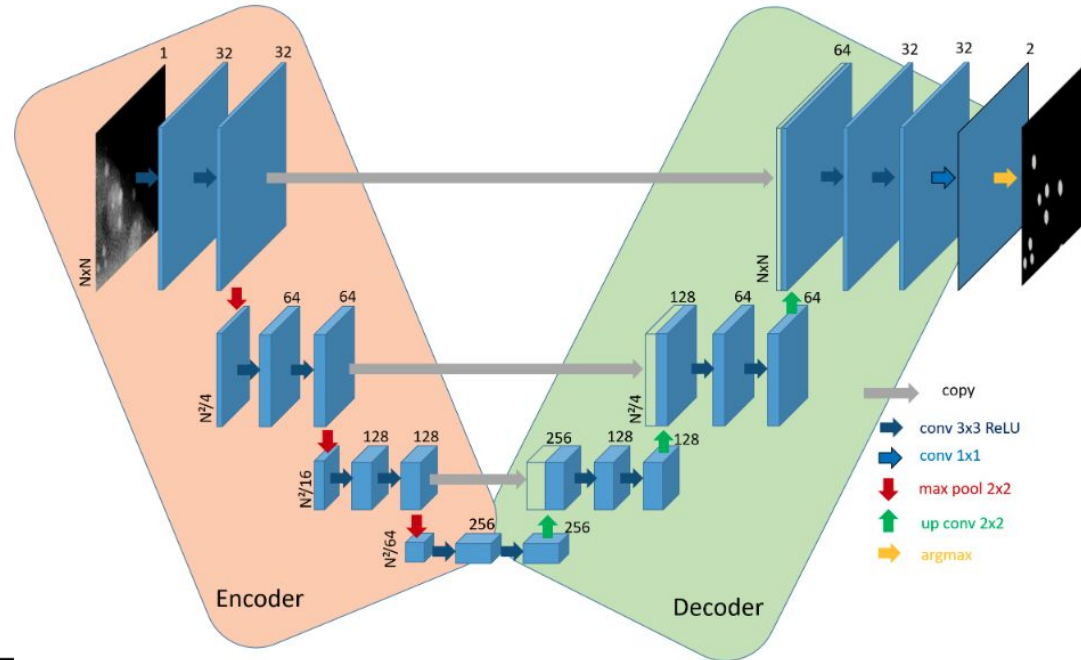
https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

b) Upload it in your online environment



In this hands-on...

You will see the code of a UNet and how it can be used to segment medical data



In this hands-on...

Start with the “test” notebook followed by the “train” notebook.

